Hot Days, High Ozone

Temperature, wind and ozone in New York, summer 1973

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Abstract

Using the daily air-quality readings that ship with R, this note asks how much of the day-to-day variation in New York's summer ozone tracks the weather. Ozone peaks in July and August, and hotter, calmer days carry markedly more of it: temperature and wind together account for well over half of the variance.

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Acknowledgements. The data are R's `airquality` set. This document exists to show a docdesigner style; its analysis is deliberately simple.

Introduction

Ground-level ozone is not emitted directly. It forms when nitrogen oxides and volatile organic compounds react in sunlight, so its concentration should follow the weather as much as it follows traffic and industry. This note checks that idea against the 153 daily readings taken in New York between May and September 1973 and published by Chambers et al. (1983), which have shipped with R ever since (R Core Team 2025).

> The greatest value of a picture is when it forces us to notice what we never expected to see.
> — John W. Tukey (1977)

Data

Each row is a day. Ozone is the mean concentration in parts per billion (ppb) from 1 to 3 pm on Roosevelt Island; solar radiation is measured in Langleys in Central Park; wind speed and the daily maximum temperature come from LaGuardia Airport. Table 1 summarises the four measures; the models below drop days with a missing reading.

Table 1: Daily measures, New York, May to September 1973

Measure	N	Mean	SD	Min	Max
Ozone (ppb)	116	42.1	33.0	1.0	168.0
Solar radiation (Ly)	146	185.9	90.1	7.0	334.0
Wind (mph)	153	10.0	3.5	1.7	20.7
Maximum temperature (°F)	153	77.9	9.5	56.0	97.0

Results

Ozone builds through the summer (Figure 1). Mean ozone in August is 60.0 ppb, roughly 2.5 times the May mean, and the highest single readings arrive in late July and August.

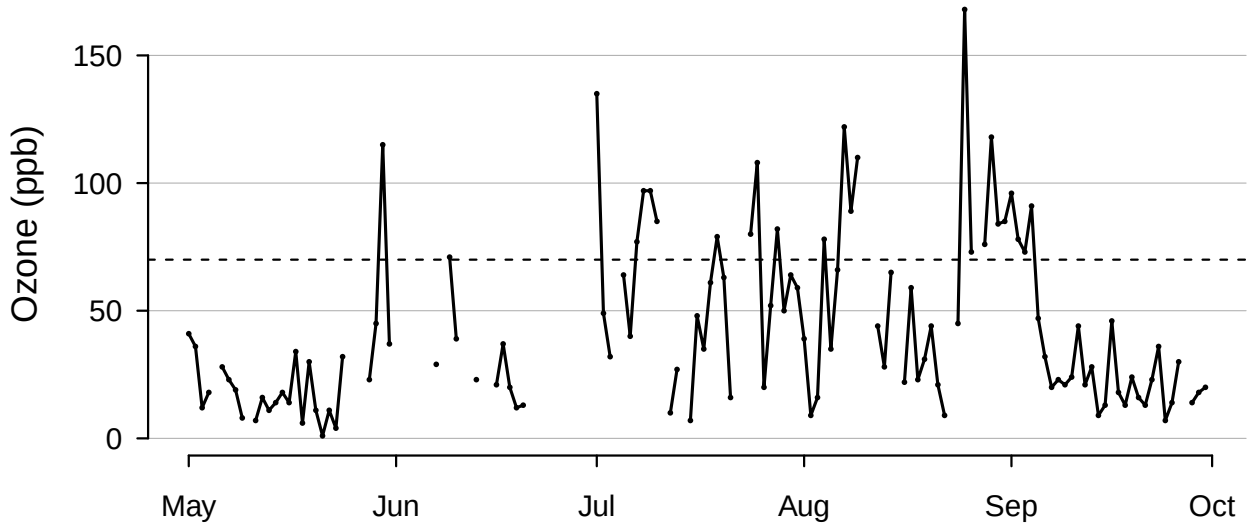

Figure 1: Daily afternoon ozone, May to September 1973. The dashed line marks 70 ppb.

Table 2 reports two ordinary least squares models. Temperature alone accounts for 49% of the variance in daily ozone; adding wind and sunlight raises that to 61%. Holding the others constant, each degree of maximum temperature goes with 1.65 ppb more ozone, and each mile per hour of wind with 3.33 ppb less: wind disperses what heat and sunlight produce.

Table 2: OLS models of daily ozone (ppb). Standard errors in parentheses; * p < .05, ** p < .01, *** p < .001

Term	Model 1	Model 2
Maximum temperature (°F)	2.43***	1.65***
	(0.23)	(0.25)
Wind (mph)		-3.33***
		(0.65)
Solar radiation (Ly)		0.06*
		(0.02)
Constant	-147.00***	-64.34**
	(18.29)	(23.05)
Observations	116	111
R^2	0.49	0.61

Conclusion

Summer ozone in New York in 1973 tracked the weather closely: the season peaked in July and August, and hot, still days carried the most ozone. None of this would surprise an atmospheric chemist, which is the point of a specimen. The reader can attend to how the page sets the argument rather than to the argument itself.

References

Chambers, John M., William S. Cleveland, Beat Kleiner, and Paul A. Tukey. 1983. *Graphical Methods for Data Analysis*. Wadsworth.

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Tukey, John W. 1977. *Exploratory Data Analysis*. Addison-Wesley.